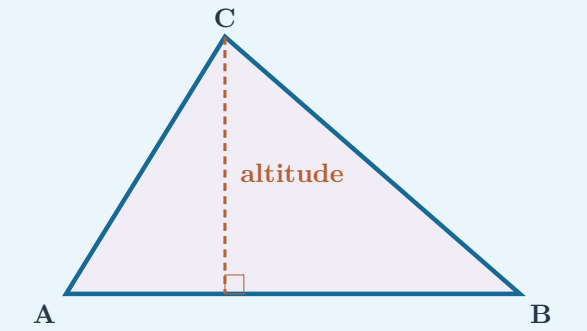
NCERT GANITA PRAKASH • CHAPTER 7

CLASS 7 MATHEMATICS • PART 1

A TALE OF THREE INTERSECTING LINES

Triangles, side lengths, angles, compass constructions and altitudes



Three sides • Three vertices • Three angles

How to use these notes

Textbook *Figure it Out* answers come first. After that: careful compass drawings, angle rules, memory tricks and a mini test. All diagrams are editable TikZ code.

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NCERT questions and complete worked answers

Navigation

The textbook has several separate *Figure it Out* groups. The heading for each group identifies its original textbook page, so repeated Q1 numbers are not confused. Constructions are described as reproducible compass/ruler steps rather than pretending one fixed drawing is the only answer.

Textbook p. 150–151: circles and isosceles triangles

Figure it Out A: Q1 • One circle and a centre

Solution. Pick **any two points** P, Q **on a circle** and use its centre O. The triangle $\triangle OPQ$ is isosceles because $OP = OQ$ (both are radii). To make it equilateral, choose points P, Q on the circle so that PQ also equals the radius.
Reason. All radii of the same circle are equal.

Figure it Out A: Q2 • Two or three circles of equal size

Solution. For two circles of radius r with centres A, B lying on each other's circumference, choose either common intersection C. Then $AB = AC = BC = r$, so $\triangle ABC$ is equilateral. With three equal circles whose centres A, B, C are mutually r apart (as in the book's picture), joining AB, BC, CA gives an equilateral triangle; triangles formed using one centre and any two points of its circle are isosceles.
Check: Equal radii, not the appearance of the diagram, prove equal sides.

Textbook p. 150–153: initial constructions

Construct • five sets of side lengths

Solution. Use **SSS construction**: draw one side as the base; draw an arc from each endpoint whose radius equals the intended adjacent side; join the arc-intersection to both endpoints.

Set (cm)	Suggested base and arc radii (cm)
(a) $4, 4, 6$	Base 6; radii $4, 4$ (isosceles).
(b) $3, 4, 5$	Base 5; radii $3, 4$ (right triangle).
(c) $1, 5, 5$	Base 1; radii $5, 5$ (isosceles).
(d) $4, 6, 8$	Base 8; radii $4, 6$ (scalene).
(e) $3.5, 3.5, 3.5$	Base 3.5; radii $3.5, 3.5$ (equilateral).

Non-examples: $3, 4, 8$ cannot form a triangle ($3 + 4 < 8$); neither can $2, 3, 6$ ($2 + 3 < 6$), $10, 15, 30$, or $3, 3, 7$.

Textbook p. 154: triangle inequality exploration

Figure it Out B: Q1–Q3 • Can these sides make triangles?

Q1: $(3, 4, 8)$: no because $3 + 4 = 7 < 8$. $(2, 3, 6)$: no because $2 + 3 = 5 < 6$.
Q2: (a) $10 + 10 = 20 < 25$ km: **no**. (b) $5 + 10 = 15 < 20$ mm: **no**. (c) $12 + 20 = 32 < 40$ cm: **no**.
Q3: Yes, for any three positive lengths in increasing order $a \le b \le c$, both $a < b + c$ and $b < a + c$ automatically hold. Only the **largest length** needs to be checked: $a + b > c$. If equal, the points lie in a straight line, not a triangle. This is the **triangle inequality shortcut**.

Textbook p. 156: seven sets of lengths

Figure it Out C: Q1 • Test every set

Case	Lengths	Triangle?	Compare smaller two with largest
(a)	$2, 2, 5$	No	$2 + 2 = 4 < 5$
(b)	$3, 4, 6$	Yes	$3 + 4 = 7 > 6$
(c)	$2, 4, 8$	No	$2 + 4 = 6 < 8$
(d)	$5, 5, 8$	Yes	$5 + 5 = 10 > 8$
(e)	$10, 20, 25$	Yes	$10 + 20 = 30 > 25$
(f)	$10, 20, 35$	No	$10 + 20 = 30 < 35$
(g)	$24, 26, 28$	Yes	$24 + 26 = 50 > 28$

Textbook p. 159: more length tests

Figure it Out D: Q1–Q3 • Equilateral and missing sides

Q1: (a) $1, 100, 100$: **yes** $(1 + 100 > 100)$. (b) $3, 6, 9$: **no** $(3 + 6 = 9)$. (c) $1, 1, 5$: **no** $(1 + 1 < 5)$. (d) $5, 10, 12$: **yes** $(5 + 10 > 12)$.
Q2: $50, 50, 50$ is possible. In general, an equilateral triangle exists for **every positive side length** s, because $s + s > s$.
Q3: For two fixed lengths a, b, the third x must satisfy $|a - b| < x < a + b$. Five examples each:
(a) $1, 100$: $99.2, 99.5, 100, 100.5, 100.8$;
(b) $5, 5$: $1, 2, 3, 4, 6$;
(c) $3, 7$: $4.5, 5, 6, 7, 8$.

Textbook pp. 160–163: SAS and ASA constructions

Figure it Out E: p. 161 • Two sides with the included angle (SAS)

Solution. Construct each by drawing the first side as a base, making the prescribed included angle with a protractor, measuring the second side along that ray, and joining the free endpoints:
(a) 3 cm, 75°, 7 cm; (b) 6 cm, 25°, 3 cm; (c) 3 cm, 120°, 8 cm. **All three are possible.**
Reason. Any *two positive* side lengths and any angle strictly between 0° and 180° determine a (non-degenerate) triangle.

Figure it Out F: p. 162 • Two angles and the included side (ASA)

Solution. Draw the given side, construct the two indicated interior angles at its endpoints, and extend the rays until they meet:
(a) 75°, 5 cm,75°: **yes**, third angle 30°.
(b) 25°, 3 cm,60°: **yes**, third angle 95°.
(c) 120°, 6 cm,30°: **yes**, third angle 30°.
When impossible? The two prescribed interior angles add to 180° or more, e.g., $100° + 80°$ or $95° + 100°$. With one angle 40°, the second must be **at least** 140° for the rays not to form a triangle.

Figure it Out G: p. 163 • Possible and impossible pairs

Q1: Choose *two possible* and *two impossible* partner angles:

Given angle	Possible partner angles	Impossible partner angles
(a) $30°$	$40°, 100°$	$150°, 160°$
(b) $70°$	$30°, 80°$	$110°, 125°$
(c) $54°$	$25°, 100°$	$126°, 130°$
(d) $144°$	$10°, 30°$	$36°, 60°$

Q2: (a) $35° + 150° = 185°$: no; (b) $70° + 30° = 100°$: yes; (c) $90° + 85° = 175°$: yes; (d) $50° + 150° = 200°$: no.
Rule: Two positive interior angles can belong to a triangle exactly when their sum is **less than** $180°$.

Textbook p. 165: angles of a triangle

Figure it Out H: Q1–Q3 • Find missing angles

Q1: Third angle $= 180° - (\text{first} + \text{second})$.
(a) $36°, 72° \Rightarrow \boxed{72°}$; (b) $150°, 15° \Rightarrow \boxed{15°}$; (c) $90°, 30° \Rightarrow \boxed{60°}$; (d) $75°, 45° \Rightarrow \boxed{60°}$.
Q2: Three angles of $70°$ are impossible since $210° \neq 180°$. With two angles $70°$, the third is $40°$. If all three are equal, each is $\boxed{60°}$.
Q3: If $\angle A = 50°$ and $\angle B = \angle C$, then $\angle B = \angle C = (180° - 50°)/2 = \boxed{65°}$.

Textbook pp. 167–171: exterior angles and altitudes

Worked textbook example • Exterior angle

Solution. If $\angle A = 50°$ and $\angle B = 60°$, the exterior angle at C is $50° + 60° = \boxed{110°}$.
Reason. Exterior angle = sum of the *two remote interior angles*. It is supplementary to the adjacent interior angle ($70°$).

Figure it Out I: p. 170 • Construct altitudes

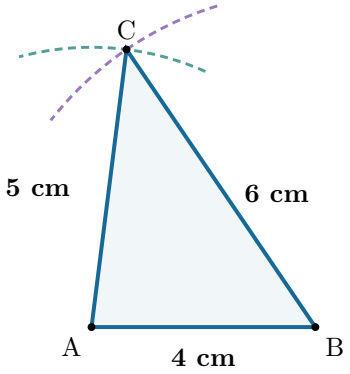
Q1: $BC = 5$ cm, $AB = 6$ cm, $CA = 5$ cm. Construct BC; intersect an arc of radius 6 centred at B with an arc of radius 5 centred at C to locate A. Join AB, AC. Draw a perpendicular from A to BC; the segment up to foot D is altitude AD. Here the foot lies on BC.
Q2: $RY = 4$ cm, $TR = 7$ cm and $\angle R = 140°$. Draw RY; construct a ray from R making $140°$ with RY and locate T at 7 cm along it. Join TY. Extend the **line** RY beyond R and draw a perpendicular from T to the extended line. The altitude foot is *outside* segment RY because the angle at R is obtuse.

Try This: p. 171 • Right and isosceles triangles

Q3: $\angle B = 90°$ and $AC = 5$ cm give **infinitely many** possible right triangles: choose any acute angle $\angle A$ between $0°$ and $90°$ and set $\angle C = 90° - \angle A$.
Q4: An equilateral triangle **cannot** be right-angled or obtuse-angled: each angle is $60°$. An isosceles triangle **can** be right-angled (angles $45°, 45°, 90°$) or obtuse-angled (angles $40°, 40°, 100°$).

Visual notes: learn the rules by drawing

1. Compass and ruler: the SSS construction

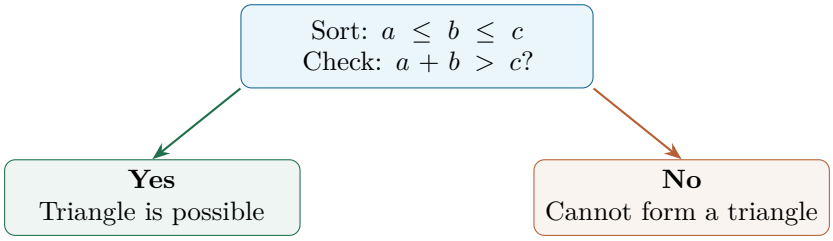


Four steps to reconstruct any SSS triangle

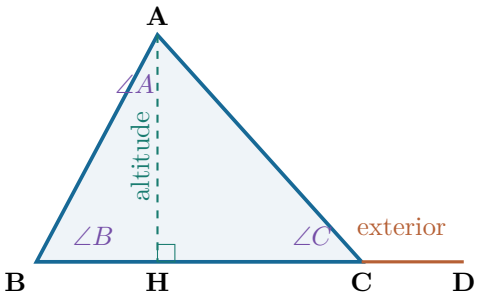
1. Draw one side. 2. Set the compass to the second side, draw an arc from one endpoint. 3. Set the compass to the third side, draw an arc from the other endpoint. 4. Join the intersection to the base endpoints.

Check existence before drawing: sum of two shorter sides must exceed the longest.

2. The triangle-inequality decision strip



3. Interior angles, exterior angle and altitude

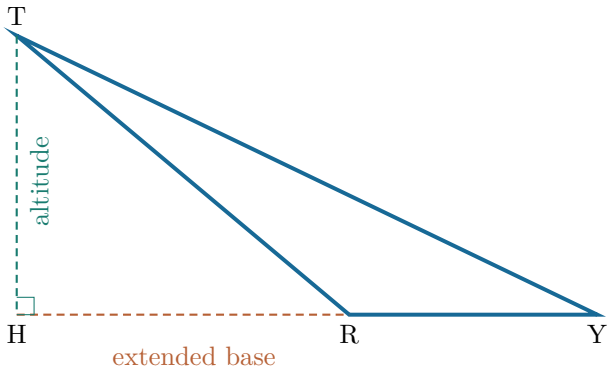


Two equations worth memorising

$$\boxed{\angle A + \angle B + \angle C = 180^\circ} \qquad \boxed{\angle ACD = \angle A + \angle B}$$

Here D lies on the extension of BC, and $AH \perp BC$. Angles are **measured at their own vertices**; the altitude is **perpendicular**, not just any sloping segment.

4. What an obtuse triangle's altitude looks like



Important: The perpendicular foot of an altitude can lie *outside* the triangle in an obtuse triangle.

Exam toolkit and common mistakes

Common mistake

- $3 + 6 = 9$ gives **no** triangle with sides $3, 6, 9$: equality is not enough.
- Do not confuse **SAS** (two sides and included angle) with **ASA** (two angles and included side).
- Two angles whose sum is **exactly** $180°$ leave a zero third angle: no triangle.
- An altitude may land **outside** a triangle; it must make a $90°$ angle with the opposite side's line.

One-minute recall

Sides: shorter + shorter > longest. **Angles:** all three sum to $180°$.
Outside: exterior equals the two remote interior angles.
Altitude: vertex to opposite side's *line* at $90°$.

Self-check: try before looking

1. Do the lengths $6, 8, 15$ form a triangle? Why?
2. Two angles are $73°$ and $47°$. Find the third.
3. One exterior angle is $132°$ and a remote interior angle is $55°$. Find the other remote interior angle.
4. Can a triangle have angles $91°$ and $90°$?
5. Give an isosceles right triangle's three angles.

Self-check answers

1. No, because $6 + 8 = 14 < 15$. 2. $60°$. 3. $132° - 55° = 77°$. 4. No: $91° + 90° > 180°$. 5. $45°, 45°, 90°$.

Source: NCERT Ganita Prakash, Class 7, Part I, Chapter 7 (2025), pp. 146–171. Exercise prompts are paraphrased; calculations and editable constructions follow the original sequence.